

# Encryption or Personal Data Protection in Envision NEXT

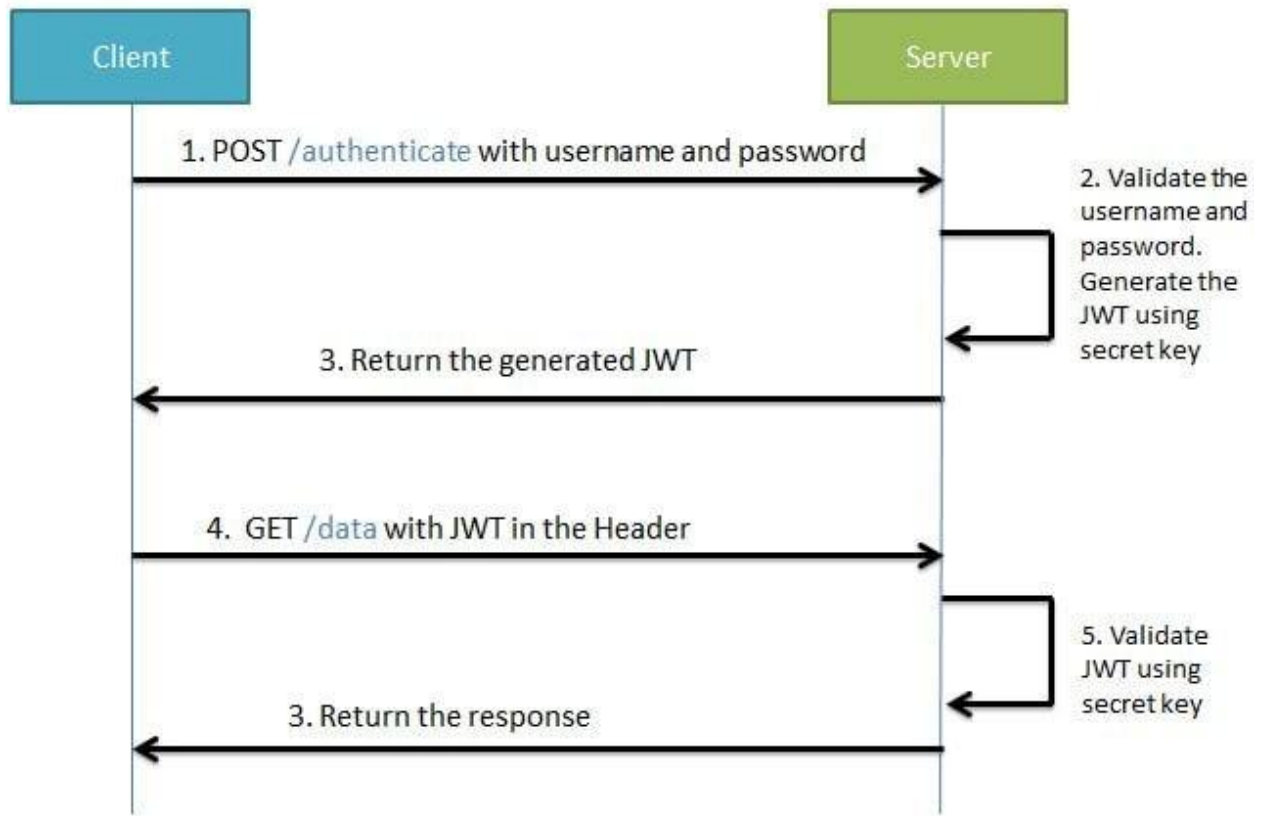
## Introduction

Database encryption converts data within a database in plain text format into meaningless cipher text using a suitable algorithm.

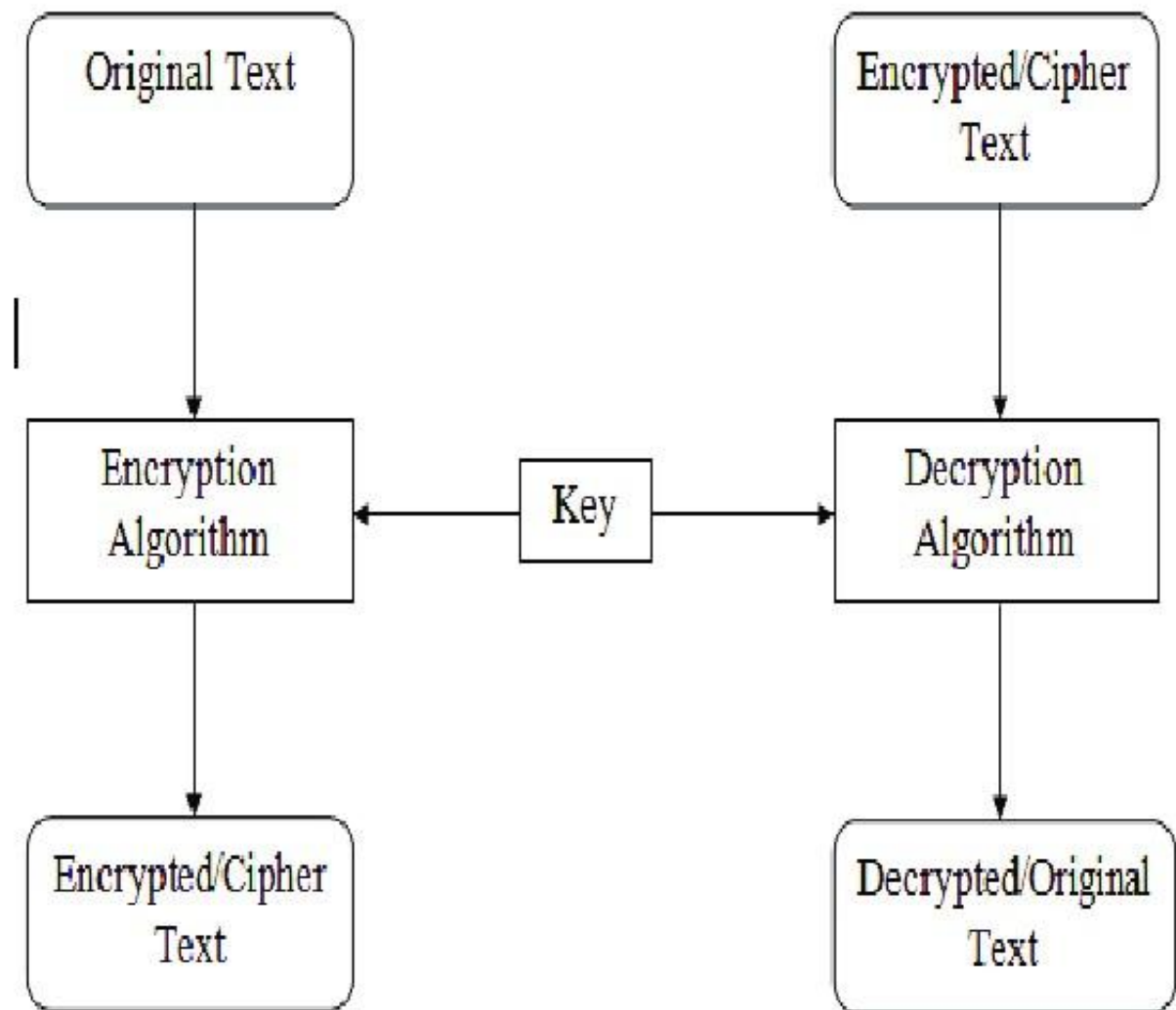
Database decryption converts the meaningless cipher text into the original information using keys generated by the encryption algorithms.

## Technology

We use JWT in Envision NEXT for Encryption or Personal Data Protection. JSON Web Token (JWT) is an open standard (RFC 7519) that defines a compact and self-contained way for securely transmitting information between parties as a JSON object. This information can be verified and trusted because it is digitally signed. JWTs can be signed using a secret (with the HMAC algorithm) or a public/private key pair using RSA or ECDSA.



## Process



# Encryption Process

**Step 1-** Taking Original Text.

**Step 2-** Apply encryption Algorithm with “**Private key.**”

**Step 3-** Taking the encrypted text.

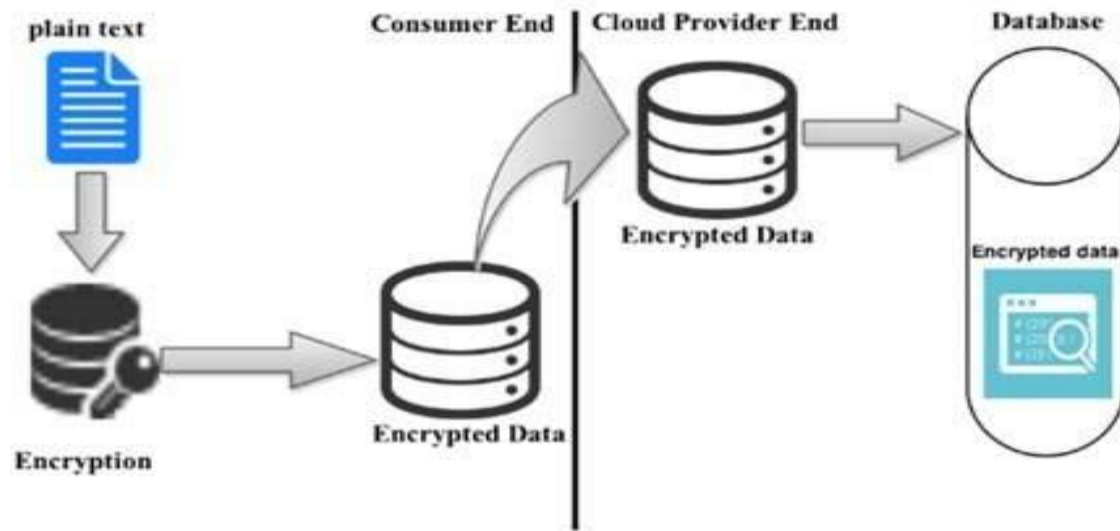
**Step 4-** Save encrypted data and database.

# Decryption Process

**Step 1-** From the database, take the encrypted text.

**Step 2-** Applying the decryption algorithm “**Private Key.**”

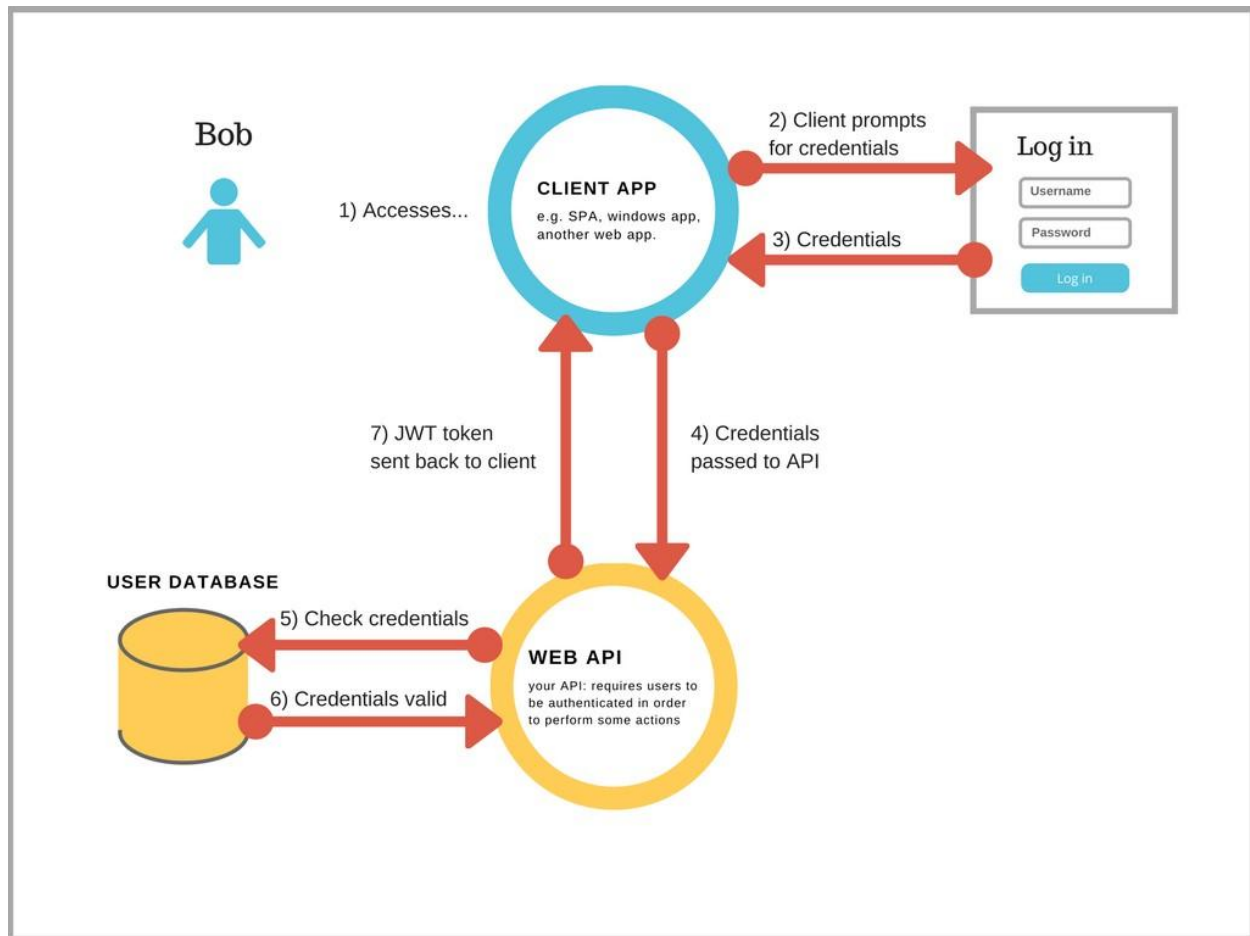
**Step 3-** Getting the original text.



## Implementation

In Envision NEXT we use this one to start with JWT, basically, all the steps would be:

- Client send a login request with username and password to server
- Server receive the username and password, authenticate the user
- If authentication is successful, then the server creates a JWT token called accessToken that stores user public info and sends it back to the client.
- Client receives the accessToken, from now on, client sends any request to server, client just attach the accessToken with request.
- Server receives a request, authenticates the JWT token, and continues processing the request, and then returns the result to the client.



## Pictorial/ Visual

**Data at rest-** When the data at rest it will fully encrypted.

**Data at travels-** The data is decrypted but the user when try to access this they have to provide JWT authentication.